Answer Key - Mathematics - Undergraduate Level Assessment 14 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) Yes, with p=2.; B) Yes, with p=3.; C) Yes, with p=5.; D) No.
   Note: Correct option: B - Yes, with p=3.

2. **Answer:** B
   Options: A) x^3 + 5 x^2 + 4 x + 1; B) x^4 - 6 x^3 + 15 x^2 - 18 x + 10; C) x^3 - x^2 + 4 x + 1; D) x^4 + 7 x^2 + 10
   Note: Correct option: B - x^4 - 6 x^3 + 15 x^2 - 18 x + 10

3. **Answer:** C
   Options: A) True, True; B) False, False; C) True, False; D) False, True
   Note: Correct option: C - True, False

4. **Answer:** C
   Options: A) 2; B) 1; C) 0; D) -1
   Note: Correct option: C - 0

5. **Answer:** A
   Options: A) f has all of its relative extrema on the line x = y.; B) f has all of its relative extrema on the parabola x = y^2.; C) f has a relative minimum at (0, 0).; D) f has an absolute minimum at (2/3, 2/3).
   Note: Correct option: A - f has all of its relative extrema on the line x = y.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'False, True'.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '1'.

9. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '2/3'.

10. **Answer:** False
    Options: A) True; B) False
    Note: LLM-generated false statement. Correct answer: 'False, False'.

Long Form Response

11. **Answer:** In an isosceles trapezoid, there are two pairs of congruent angles. Let x and y be the distinct angle measures. Since the angles of a quadrilateral sum to 360, we have $2x + 2y = 360$. Setting $x = 40$ we find $y = 140$ degrees.
    Note: Students should show all steps in their mathematical solution.

12. **Answer:** 7. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
    Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key